CO2 - AT1

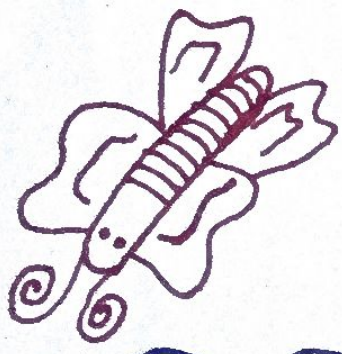
Analytical Problem Solving

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Thresholding and Intensity Level Slicing - PCB Inspection:

Given:

$$x = [25, 60, 95, 130, 165, 200, 235].$$

a) Binary thresholding:

$$S = \begin{cases} 255, & x \geq 128 \\ 0, & x < 128. \end{cases}$$

Input(x)	Output(s):
25	0
60	0
95	0
130	255
165	255
200	255
235	255.

Sol: [0, 0, 0, 255, 255, 255, 255].

b) Intensity - level slicing:

Input (x)	Output (s).
25	25
60	60
95	255
130	255
165	255
200	200
235	235.

Sol: [25, 60, 255, 255, 200, 235]

The pixels in the intensity band 90-170 set to 255. All other pixels remain

c) No. of highlighted and percentage pixel

Highlighted pixels are 95, 130 & 165.
No. of highlighted pixels = 3
Total no. of pixels = 7.

$$\text{Percentage} = \frac{3}{7} \times 100$$

$$= 42.86\%.$$

d) Mean intensity:

Original row:

$$\bar{x} = \frac{25+60+95+130+165+200+235}{7}$$

$$\bar{x} = \frac{910}{7} = 130.$$

$$\bar{x} = 130.$$

Sliced Output:

$$\bar{s} = \frac{25+60+255+255+255+200+235}{7}$$

$$\bar{s} = \frac{1285}{7} = 183.57$$

$$\bar{s} = 183.57.$$

Comparison and Justification:

Binary thresholding	Intensity level
Highlights every pixel greater than or equal to 128, including 200 & 235, which are outside the required band.	Highlights only pixels between 90 & 170, which contain the solder joints of interest.

2) Piecewise - Linear Contrast Stretching X-ray Image Enhancement:

a) Equation and Slopes of the three segments.

The straight line between two control points:

$$S = ar + b \quad ; \quad a = \frac{S_2 - S_1}{r_2 - r_1}$$

Segment 1: $0 \le r \le 70$

(0,0) & (70, 20).

$$a_1 = \frac{20 - 0}{70 - 0} = \frac{2}{7} = 0.2857.$$

$$\boxed{S = \frac{2}{7} r}$$

Segment 2: $70 \le r \le 140$

$$a_2 = \frac{230 - 20}{140 - 70} = \frac{210}{70} = 3$$

$$S = ar + b \Rightarrow 20 = 3(70) + b$$

$b = -190$

$\boxed{s = 3r - 190.}$

Segment 3: $140 \leq r \leq 255$.

(140, 230) & (255, 255).

$$a_3 = \frac{255 - 230}{255 - 140} = \frac{25}{115} = 0.2174$$

$$\Rightarrow 230 = \frac{5}{23}(140) + b.$$

$$\boxed{s = \frac{5}{23} r + \frac{4590}{23}.}$$

The Complete Transformation is:

$$S(r) = \begin{cases} \frac{2}{7} r, & 0 \leq r \leq 70 \\ 3r - 190, & 70 \leq r \leq 140 \\ \frac{5}{23} r + \frac{4590}{23}, & 140 \leq r \leq 255. \end{cases}$$

b) Transformed values:

Input pixels:

[30, 70, 90, 110, 140, 200].

Input(r)	Segment	Calculation	Output (s)
30	1	(2/7) (30)	8.57
70	1 or 2	20	20
90	2	3(90) - 190	80
110	2	3(110) - 190	140
140	2 or 3	230	230
200	3	(5/23)(200) + 4590/23	243.04

width of the diagnostic band and contrast gain:

Original band width:

$$W_{orig} = 140 - 70 = 70.$$

Trans formed band width:

$$W_{trans} = 230 - 20 = 210.$$

Contrast gain factor:

$$G_1 = \frac{W_{trans}}{W_{orig}} = \frac{210}{70} = 3.$$

The contrast of the diagnostic band is increased by a factor of 3.

d) Compression and Expansion:

Segment	Slope	Effect.
0-70	0.2857	Compresses intensity range.
70-140	3	Expands intensity range.
140-255	0.2174	Compresses intensity range.

e) Assessment:

The transformation improves the visibility the diagnostic region because the grey-level 70-140 is expanded from 70 levels to 210 le

3) Spatial Filtering:

$$\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 59 \end{bmatrix}$$

a) 3x3 mean (box) filter:

$$\text{Mean} = \frac{\text{Sum of all 9 pixels}}{9}$$

$$= \frac{52+55+58+54+255+57+53+56+59}{9}$$

$$= \frac{699}{9}.$$

Mean filter output = 77.67.

b) Weighted average filter:

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

$$S = \frac{1}{16}[52+110+58+108+4(255)+114+53+112+59].$$

$$S = 1686/16.$$

Weighted filter = 105.37

c) Median filter:

Arrange all nine value in ascending order:

52, 53, 54, 55, 56, 57, 58, 59, 255.

Median filter = 56.

d) Absolute error of each filter:

$$R = \frac{52+55+58+54+57+53+56+59}{8}$$

$$R = \frac{444}{8} = 55.5.$$

Absolute Error Formula:

$$E = |\text{Filter output} - R|.$$

$$E_{mean} = |77.67 - 55.5| = 22.17$$

$$E_{weighted} = |105.375 - 55.5| = 49.875.$$

$$E_{median} = |56 - 55.5| = 0.5.$$

4) Edge Detection Sobel:

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{bmatrix}.$$

a) Sobel Operator :

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

G :

$G_x = (-10+10) + (-20+180) + (-10+90).$

$$\boxed{G_x = 240}$$

$G_y = (-10 - 20 - 10) + (10 + 180 + 90).$

$$\boxed{G_y = 240}$$

b) Gradient :

$$M = \sqrt{G_x^2 + G_y^2} = \sqrt{240^2 + 240^2}$$

$$\boxed{M = 339.41}$$

$$M_{approx} = |G_x| + |G_y| = 240 + 240$$

$$\boxed{M_{approx} = 480.}$$

Percentage Error :

$$Error = \frac{|M_{approx} - M|}{M} \times 100.$$

$$= \frac{|480 - 339.41|}{339.41} \times 100.$$

$$\boxed{\varepsilon_{error} = 41.42\%}$$

c) Gradient direction and edge orientation:

$$\alpha = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$\alpha = \tan^{-1}\left(\frac{240}{240}\right)$$

$$\boxed{\alpha = 45^\circ}$$

The gradient direction is 45°. The edge itself is perpendicular to the gradient, so its orientation is approximately 135°, or equivalently -45°.

d) Repeat using the Prewitt operator:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$G_x = (-10+10) + (-10+90) + (-10+90)$$

$$G_x = 0 + 80 + 80$$

$$\boxed{G_x = 160}$$

G_y:

$$G_y = (-10-10-10) + (10+90+90)$$

$$G_y = -30 + 190$$

$$\boxed{G_y = 160}$$

Exact Magnitude:

$$M = \sqrt{160^2 + 160^2}$$

$$M = \sqrt{51200}$$

$$\boxed{M = 226.27}$$

Approximate magnitude:

$$M_{approx} = |160| + |160|$$

$$\boxed{M_{approx} = 320}$$

Percentage error:

$$E_{error} = \frac{|320 - 226.27|}{226.27} \times 100$$

$$\boxed{E_{error} = 41.42\%}$$

Edge detection using $M \geq 250$

Operator	Exact Magnitude	Threshold	Edge detected?
Sobel	339.41	250	Yes
Prewitt	226.27	250	No.

5) Global Thresholding - Iterative & Otsu Segmentation

Intensity g_k	0	1	2	3	4	5	6	7
Frequency n_k	3	6	4	1	1	5	8	4

Total numbers of pixels:

$$N = 3+6+4+1+1+5+8+4.$$

a) Global mean intensity:

$$m_G = \frac{\sum g_k n_k}{N}$$

$$m_G = \frac{0(3)+1(6)+2(4)+3(1)+4(1)+5(5)+6(8)+7(4)}{32}$$

$$m_G = \frac{122}{32}$$

$$\boxed{m_G = 3.8125}$$

b) Iterative global thresholding.

$$T_0 = m_G = 3.8125$$

Use the following rule:

*) pixels with $a \leq T$

*) pixels with $a > T$.

$$T_{new} = \frac{m_1 + m_2}{2}$$

Iteration 1:

$$T_0 = 3.8125$$

$$m_1 = \frac{0(3) + 1(6) + 2(4) + 3(1)}{3+6+4+1.}$$

$$m_1 = \frac{17}{14} = 1.2143.$$

$$m_2 = \frac{4(1) + 5(5) + 6(8) + 7(4)}{1 + 5 + 8 + 4.}$$

$$m_2 = \frac{105}{18} = 5.8333$$

$$T_1 = \frac{1.2143 + 5.8333}{2}$$

$$\boxed{T_1 = 3.5238}$$

Threshold change:

$$\Delta T = |3.8125 - 3.5238|$$

$$\boxed{\Delta T = 0.2887}$$

Since $0.2887 < 0.5$, Stop the Iteration.

Final Iterative threshold:

$$T = 3.5238$$

$$\boxed{T \approx 3.52}$$

c) Otsu's method: Between-class variance:

$$\sigma_B^2(k) = P_1(k) P_2(k) [m_1(k) - m_2(k)]^2$$

For $K = 2$:

Class 1: levels 0, 1, 2:

$$P_1 = \frac{13}{32}, \quad m_1 = \frac{14}{13} = 1.0769$$

Class 2: levels 3, 4, 5, 6, 7

$$P_2 = \frac{19}{32}, \quad m_2 = \frac{108}{19} = 5.6842$$

$$\sigma_B^2(2) = \frac{13}{32} \frac{19}{32} (1.0769 - 5.6842)^2$$

$$\boxed{\sigma_B^2(2) = 5.119}$$

For K=3:

Class 1 : levels 0, 1, 2, 3

$$P_1 = \frac{14}{32}, \quad m_1 = \frac{17}{14} = 1.2143$$

Class 2 : levels 4, 5, 6, 7.

$$P_2 = \frac{18}{32}, \quad m_2 = \frac{105}{18} = 5.8333$$

$$\sigma_B^2(3) = \frac{14}{32}\,\frac{18}{32}\,(1.2143 - 5.8333)^2$$

$$\boxed{\sigma_B^2(3) = 5.250}$$

For K=4:

Class 1 : level 0, 1, 2, 3, 4

$$P_1 = \frac{15}{32}, \quad m_1 = \frac{21}{15} = 1.4$$

Class 2 : level 5, 6, 7

$$P_2 = \frac{17}{32}, \quad m_2 = \frac{101}{17} = 5.9412$$

$$\sigma_B^2(4) = \frac{15}{32}\,\frac{17}{32}\,(1.4 - 5.9412)^2$$

$$\boxed{\sigma_B^2(4) = 5.137}$$

Optimum threshold and binary mapping

Otsu's method selects the threshold with the maximum between class variance.

Since 5.850 is the largest value:

$$k^* = 3$$

Percentage of pixels classified as objects:

object pixels $= 3+6+4+1 = 14$

object percentage $= \frac{14}{32} \times 100$

object percentage $= 43.75\%$

e) Compare the two methods:

Method	Threshold	Result
Iterative global thresholding	3.5238	Separate levels 0-3 from 4-7.
Otsu's method	3	Separate levels 0-3 from 4-7.